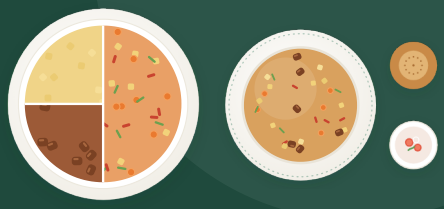
UNIVABIO 2026 · ARTIFICIAL INTELLIGENCE FOR HUMAN HEALTH

FoodWise

Eat Well. Stay You.

An AI-powered nutrition **education** assistant. It suggests practical, culturally familiar meals for each person's real situation — budget, time, hunger, allergies, traditions — and explains *why* they work, so people gradually learn to make balanced choices on their own.

▶ Demo: aaegamova-code.github.io/foodwise | GitHub: aaegamova-code/foodwise



The problem

Most nutrition apps are calorie trackers or rigid meal plans. They rarely fit local food cultures, ignore everyday constraints like a 15-minute lunch break or a tight budget, and frame traditional dishes as "unhealthy". Users follow instructions without understanding them — and drop out.

In Uzbekistan, daily meals revolve around beloved dishes like plov, shurpa and non — people need guidance that **works with** these foods, not against them.

Our solution — three pillars

1. **Context-Aware AI.** Combines stable context (allergies, diet, culture, goals) with today's situation (skipped meals, fatigue, budget in UZS, time, place) to suggest 2–3 realistic meals.
2. **Evidence-based education.** Every suggestion has a "Why this works" explanation linked to WHO healthy-diet principles and a visual plate composition.
3. **Progressive Independence.** Short, judgement-free exercises teach users to reason about meals themselves, reducing reliance on the app over time.

Five Principles. One Smarter Choice.

Four principles from WHO/FAO healthy-diet guidance + Satisfaction, FoodWise's own proposed addition (not an official WHO principle).

WHO	WHO	WHO	WHO	FOODWISE
Adequacy	**Balance**	**Moderation**	**Diversity**	**Satisfaction**
Enough of what the body needs, without excess.	A sensible mix of energy and macronutrients.	Limiting excess sugar, salt and some fats.	Variety across food groups.	Taste, tradition and practicality — so habits last.

User journey (demonstrated in the prototype)

1. **Share context** — Free-text profile; allergies confirmed explicitly
2. **Describe today** — "Skipped breakfast, 50,000 UZS, 15 min"
3. **Get options** — 2–3 meals with plate visuals, cost, time
4. **Understand why** — Reasons, principles, practical tweak
5. **Practise** — Choose a combination, get constructive feedback
6. ★ **Choose alone** — Variety? Enough? Enjoyable?

Health impact & scientific rigour

- Promotes balanced, varied everyday eating while keeping cultural foods — plov is adjusted, not banned.
- Confirmed allergies and dietary exclusions are enforced in **every** suggestion, follow-up and swap.
- No diagnosis or guaranteed outcomes; medical conditions, emergencies and extreme dieting are redirected to professionals.
- Real sources (WHO, FAO, Harvard), labelled apart from clinical (IFM) and coaching (IIN) perspectives.
- Privacy by design: all personal and health data stays on the user's device.

Technical implementation

- React + TypeScript + Vite, mobile-first (375–430 px), deployed on GitHub Pages.
- **Interactive Demo engine**: transparent rule-based recommender over 18 structured meals; handles budget, time, diet, dislikes, swaps and requested dishes.
- Structured responses ready for a server-side LLM; the allergy filter stays as a final safety layer.
- Dynamic SVG plate & bowl illustrations: share of plate area (not calories), inspired by the Harvard Healthy Eating Plate.

Working now

- All 4 screens, context setup & editing
- Recommendations, "Why this works", swaps
- Progressive Independence exercise

Simulated in the demo

- AI reasoning is rule-based (labelled "Interactive Demo")
- Prices are illustrative UZS estimates, not live data
- No clinical validation yet

Next steps

- Secure server-side LLM integration
- Review by dietitians & physicians
- Uzbek & Russian UI, local price data
- Pilot study with students